

Digital Essay Assessment using Machine Learning Algorithms

R. Johnsi and G. Bharadwaja Kumar

School of Computer Science and Engineering, Vellore Institute of Technology, Chennai.

E-mail : johnsi.2020@vitstudent.ac.in, bharadwaja.kumar@vit.ac.in

Abstract- Essay are considered as the most prominent way in assessing the student performance. Manual grading of the essay are considered to be a hectic tasks for both students and instructors due to various reasons such as time consuming, prone to subjectivity, inconsistencies and lack of reliability may be influenced by biases, such as prior knowledge of the student or their back ground. The solution to all these problems has been rectified by using Automated Essay Scoring (AES) system. This technical paper proposes a novel AES system that utilizes various Machine Learning (ML) algorithms such as Linear Regression, Lasso Regression, Ridge Regression, Gradient Boosting Regression, Bayesian Regression, Elastic Net Regression, Random Forest Regression, Decision Tree Regression, Gaussian Process Regression, Support Vector Regression, Polynomial Regression, XG Boost regression and XGBoost with Polynomial Features and discovered that XGBoost with Polynomial features achieved the best results with an average Quadratic Weighted Kappa (QWK) value of 0.68.

Keywords—Computer-based assessment system, Essay scoring, Natural Language Processing, Semantics, Linguistics, Education, Text Processing.

I. INTRODUCTION

Global reaction to the CORonaVirus Disease (COVID-19) pandemic has been significantly helped by the World Health Organization (WHO) (<https://covid19.who.int/>). As the leading international agency for public health, the WHO has played a key role in coordinating international efforts to control the spread of the virus and mitigate its impact. Since its outbreak in late December 2019, WHO declared COVID-19 to be a pandemic on 11th March 2020. The COVID-19 pandemic has a significant impact on education systems worldwide. The closure of schools and universities to prevent the spread of the virus has disrupted learning for millions of students and forced educators to quickly adapt to new online teaching methods. So, educators have quickly adapted to online learning platforms to continue delivering education remotely. Thus, the pandemic has accelerated the adoption of online education and highlighted the importance of digital literacy and teacher training in online learning [1].

Despite of online and offline learning process, assessment is an integral part in all educational system.

The goal of assessments of learning progress is to evaluate student progress towards learning goals and to identify areas where students need additional support or instructions [2]. In both traditional and online education settings, assessment scoring is mostly done by manual or human grading. This process involves teachers or graders reviewing and evaluating student assignments, and assigning scores or grades based on predetermined criteria. But it faces some limitations such as time consuming, prone to subjectivity, inconsistencies, lack of reliability and may be influenced by biases, such as prior knowledge of the student or their back ground.

To overcome to above limitations, manual scoring is made as substitute named as AES system. It is a technology that uses computer algorithms to evaluate and score essays without human involvement [3]. The use of AES has increased in schools and universities have moved towards remote learning and assessment during the COVID-19 pandemic [4]. AES provides a standardized approach to grading essays, reducing variability and ensuring consistency in grading across different assessors. It can save time for teachers or educators who would otherwise have to read and grade essays manually, allowing them to focus on other aspects of teaching. The system can increase access to education by allowing for automated grading of large numbers of essays, making it possible to grade more essays in less time. It can reduce the potential for bias in grading by removing unwanted information from essays and evaluating them based solely on their content [5].

Despite its benefits, AES does not come without obstacles. Some critics have raised concerns about the accuracy of AES, particularly in assessing essays with more complex language and arguments. But many researchers was overthrown to mitigate these limitations in near future. Using these actuations, we also explore the potential relevance of lexical features in enhancing AES accuracy using various ML models.

Motivation

The main motivation, to have an AES task is due to the following reasons:

1. The subjective nature of essay evaluations results in error and variations in scores and awarded by

human assessors, which is perceived by students as a source of unfairness. By having the AES tasks can overcome the limitation of reliability, human bias etc. [6].

2. According to the Right of Children to Free and Compulsory Education (RTE) Act, 2009, the student -Teacher Ratio (PTR) norms for Government upper primary schools is 24:1 as stated [7]. If there is disparity, will lead to poor academic performance of students. The huge amount of time spent on evaluation can be utilized for improving teaching learning process [8].
3. Different type of answers will be given by student for a single question (length, texts). It varies in the terms of the length of text, language modulation, and grammatical point of views [9].
4. Due to some personal reason or mood swings of teachers will leads to different scores same assessor in different times [10].

In order to overcome these above limitations, the proposed research aim is to develop an AES system that utilizes various ML methodologies. But still there is a lack in developing a robust AES tasks.

A lot of the work in essay grading suggest that millions of assessments are conducted worldwide each day. Manual grading these assessments is tedious and subjective. Instead, many organizations have already turned to automated essay grading to improve consistency, reliability saves time and resources, provides immediate feedback, AES can be customized to evaluate specific aspects of writing, such as grammar, syntax, and content, allowing educators to tailor assessments to meet specific learning objectives. AES is an innovative and cutting-edge approach to evaluating student writing, which can help educators stay up-to-date with advances in educational technology and teaching methods.

The subsequent sections of this paper are structured as follows. In Section 2, presents a historical database and offers an overview of related work in the literature. Section 3 outlines the general framework of the proposed AES system. Section 4 serves as the conclusion and future work of the paper.

II. HISTORICAL BACKGROUND AES (AES) SYSTEM

AES was primarily utilized to overcome time and expense, and reliability, subjective nature of the essay, validity and generalizability issues in writing assessment. The limitation and necessity for automating the evaluation process has become the demand of the hour. A number of studies has been made to replicate the limitation of AES system.

The history of AES dates back to the 1960s, when researchers first began exploring the use of computers for language processing tasks. However, it was not until the 1990s that AES technology became widely available and began to be used in educational and testing contexts.

The first developed AES system was Project Essay Grade (PEG), developed by Ellis Batten Page in the 1960s. PEG used a set of rules and algorithms to evaluate essays based on factors such as grammar, syntax, and vocabulary. Ellis Batten use the Multiple Linear Regression (MLR) as statistical techniques to identify patterns in the data and determine a score for the essay being evaluated and acquired the correlation value of 0.80 [11]. As PEG system used only the statistical features, thus it focus on the writing style rather than the content based features. However, the use of technology of that time, desktop computers had not become so powerful and so, widespread that AES was a not practical possibility and would have been cost-effective. Thus Ellis Batten Page sold PEG to measurement incorporated (<https://measurementinc.com/>) and continued to develop it.

Peter Foltz and Thomas Landauer developed the Intelligent Essay Assessor (IEA) as a continuation of PEG system. The system was first commercial used to score an essay in 1998. The system has been developed based on considering features such as content, style, plagiarism, mechanics and validations [7]. IEA analyzed the content based features using Latent Semantic Analysis (LSA). LSA analyses the semantic relationship between the words and texts and acquired the correlation value of 0.90 between IEA and the human-raters. The system also supplies immediate feedback on both content based and the quality of the student's writing.

Then E-rater developed the Educational Testing Services (ETS) system (<https://www.ets.org/erater.html>) by using combination of both statistical and NLP based features. ETS system was first used commercially in 1999. The E-rater system uses 11 features derived from both writing quality and content based aspects. Following these, the weights for the features are determined using a multiple regression procedure. Performance are computed and the correlation between human assessors and the system ranged from 0.87 to 0.94.

Criterion is an essay evaluation system created by ETS and accessible through the web. The system uses AES technology to evaluate essays based on a variety of factors, including content, organization, and language use. Besides providing instant holistic scoring, criterion also provide diagnostic instant feedback to students, allowing them to revise and improve their essays before submitting a final version.

IntelliMetric systems was developed by vantage learning in year 1998 [8]. The system used the combination of technology such as Artificial Intelligence (AI), NLP, and statistical technologies. IntelliMetric takes into account various factors when scoring an essay, including grammar, vocabulary, organization, coherence, and argumentation. The system provides the essay score based on the features such as focus, organization, content, language, style mechanics and conventions. The average correlation value ranges between automated and human-raters is 0.83.

MY Access is an online writing program that developed in late 1990. It provides students with immediate feedback and guidance on their writing skills. The program uses advanced algorithms and NLP technology to analyze and evaluate student writing, providing feedback on areas such as grammar, syntax, vocabulary, and organization. MY Access also provides students with individualized instruction and practice exercises to help them improve their writing skills over time.

Bayesian Essay Test Scoring System (BETSY) was developed in 2002. It is a ML algorithm that is used to score essays and other written assignments. It is based on Bayesian statistics, a type of statistical analysis that uses probability to make predictions and draw conclusions. BETSY analyzes and Natural Language Processing (NLP) techniques. BETSY has attained more than 80% accuracy when compared automated score with human score.

In continuation of period after 2000 has termed as a transformation in AES research task through integration of many ML algorithms. These advancements have led to more efficient, accurate and scalable AES system and making a significant impact on educational assessment practices.

III. RELATED WORK

Automated Essay Grading Using Machine Learning:

In this proposed work, [9] author uses the ASAP (Automated Student Assessment Prize) dataset and extracted features such as Bag of Words (BOW), numerical features, Parts-Of-Speech (POS) count, orthography, structure and organization. Then with these extracted features a simple LR model is used to get the final machine score. The performance was tested using QWK score and achieved score of 0.73. The main drawback exists in this work is that it does not well for narrative type of an essay. Thus the system has to been enhanced by using the advanced NLP features such as n-gram model.

Building e-rater® Scoring Models Using Machine Learning Methods

In current usage, e-rater was deployed by extracting the

various features from each essay set and final scores are generated by using MLR algorithm. Thus in this proposed work, alternative scoring models are deployed by comparing using various ML algorithms such as K-Nearest Neighbor regression (k-NN), Support Vector Machine (SVM), and Random Forests (RF) [10]. The dataset used is collected from two large-scale college level assessments. Thus the results suggest that the correlation between the human score and machine score is less by using SVM algorithm. The author concludes by highlighting that AES can also be viewed as the classification problem, aiming to predict the categorical score of an essay in order to enhance the performance of an essay scoring process.

Exploring the potential of using an AI language model for AES

The proposed work was implemented in order to explore the importance of chat Generative Pre-trained Transformer (GPT). The use of GPT along with the additional features helps in improving the performance of an AES system. The author used the ETS Corpus. It consists of total collections of 12,100 english essays written by non-native english learners [11].

Automated Essay Scoring Using Machine Learning

The author proposed the research work by extracting the five different types of features which is based on the visual nature, inclusion of pronouns, beautiful words, emotion effectiveness, maturity of words [12]. Based on these extracted features, AES as a regression problems is deployed by using various algorithms such as LR, SVM, and multi-class Linear Discriminate Analysis (LDA) and Regression Trees (RTs). The performance was tested by using QWK as metrics and suggest that SVM outperforms other ML models with an average QWK values of 0.78.

GENERAL FRAMEWORK OF AES SYSTEM

The framework of an AES system typically involves several key components working together to evaluate and score essays.

The general framework for an AES system includes,

- Data gathering
- Pre-processing
- Feature extraction
- Scoring model
- Evaluation metrics

Data gathering:

There are many dataset used in implementation of AES system in past decade. But mostly 90 % AES work is based on ASAP dataset. It is founded by Hewlett Foundation's (<https://www.kaggle.com/c/asap-aes>) which is a publicly accessible dataset containing essays that have been manually scored and used to train and evaluate AES systems. The Hewlett foundation provided the dataset in 2012 as part of the ASAP competition, which intended to

foster the development of accurate and trustworthy AES systems. It consists of 12,978 samples of essays under eight set of essays. The score of an each essay varies from 0-60. The essays in the dataset are diverse and include expository, argumentative, and narrative essays written by students in grades 7-12. The dataset includes essays on a wide range of topics such as technology, education, social issues, and more. Table 1 describes the description of ASAP data.

Pre-processing

The essay goes through a preprocessing stage where it is prepared for analysis. This step may involve removing irrelevant information, normalizing the text, and handling any formatting or structural issues. The basic preprocessing steps done is tokenization, removing stop words, stemming etc.

Feature extraction

Feature extraction is another crucial stage in the building

of an AES system. The raw essay text is converted into a set of numerical characteristics that may be utilized as inputs to the ML algorithm. In this, author proposed the automated essay grading tasks as a regression problem. Here, 12 features are extracted based on coherence, essay writing pattern, cohesion and vocabulary uniqueness in each essay set and scores are finalized by using XGBoost classifier and acquired average accuracy of 66.87% [13].

Thus contagious of the above paper, we proposed AES tasks as a regression problem by extracted 12 features from each essay and yields the final score. These features are based on lexical and syntactic characteristics, such as word frequency, sentence length, grammatical structures, and cohesive to extracts various features from the essay that can be used to assess its quality. Python was utilized for implementation part and achieves the average QWK values as 0.68. The following explains the each features in a detailed manner.

TABLE 1 ASAP DATASET DESCRIPTIONS

Essay set	Number of essays	Type of Essay	Average Length of words	Maximum mark of essay	Student Response
1	1783	Persuasive/ Narrative/ Expository	350	12	Grade 8
2	1800	Persuasive/ Narrative/ Expository	350	6	Grade 10
3	1726	Source Dependent Responses	150	3	Grade 10
4	1772	Source Dependent Responses	150	3	Grade 10
5	1805	Source Dependent Responses	150	4	Grade 8
6	1800	Source Dependent Responses	150	4	Grade 10
7	1569	Persuasive/ Narrative/ Expository	250	30	Grade 7
8	723	Persuasive/ Narrative/ Expository	650	60	Grade 10
Total	12978				

1) Word count: The number of words used is considered as an important criteria for essay writing. This is obtained by splitting each sentences into individual words as a tokens.

2) Sentence count: The total number of sentences in an essay is counted to determine it.

3) Average of Sentence length: This is computed by taking the essay's overall sentence length and dividing it by the total number of sentences.

4) Comma count: This is accomplished by calculating the total amount of commas in an essay using the spacy toolkit.

5) POS tag : POS tagging is a common task in NLP that involves identifying the grammatical category of each word in a sentence, such as noun, verb, adjective, etc. It

helps to understand the grammatical structure and meaning of text in essays. In this proposed work, POS was tracked using popular python Natural Language Toolkit (NLTK).

6) Out Of Vocabulary (OOV): OOV words are words that do not appear in the language model's training data or vocabulary, and they can have an impact on the AES system's accuracy and dependability.

7) Parse tree depth: It refers to the number of layers or levels in a syntactic tree that represents the grammatical structure of a sentence. This work [14] explains how to compute the depth of parse tree in a sentences.

8) Formality: Formality features helps to better understand the level of formality in a given text and to adapt the language use accordingly. In NLP, formality

features refer to linguistic cues that can help identify the level of formality in a given text or speech [15]. The following equations (1) helps to analyze the structure of sentences.

$$F = \frac{\sum_{i \in A}^n \left(\frac{c(i)}{n}\right) \sum_{j \in B}^n \left(\frac{c(j)}{n}\right) + 100}{2} \quad (1)$$

Where $A = \{N, DET, PP, ADJ\}$

$B = \{V, ADV, PR, UH\}$ and n denotes the number of tokens in the text.

9) Sentence similarity percentage: This is accomplished by adding up the results of similarity comparisons between each essay sentence along with other sentences, making use of the spacy toolkit.

10). Noun count percentage in every single sentence: Noun count percentage in each sentence refers to the amount or percentage of the nouns in a sentence in proportion to the total number of words. A higher noun count percentage may indicate that a sentence is more focused on describing objects or concepts, while a lower noun count percentage may indicate a more abstract or subjective sentence. To calculate the noun count percentage in a sentence, the total number of nouns in the sentence is divided by the total number of words in the sentence and multiplied by 100 to express the result as a percentage.

11) Transition word percentage: It is calculated by dividing the results of the transition word check by the total count of transition words in a set. The outcome is then multiplied by 100.

12) Type Token Ratio (TTR): It is calculated from dividing the number of unique words with the total number of words in the essay. It will then be multiplied by 100. Type-token-ratio to estimate whether an essay has a relatively rich or rather poor vocabulary.

Scoring model

Once features are extracted based on four categories from each essay set. Then these extracted features are sent into ML models to get a single final score as an output. AES can be framed as a regression problem. The goal of AES is to predict a score for an essay based on its content, structure, and other features. This score can be a continuous value, such as a score on a 0-60 scale, or a discrete value, such as a grade or a pass or fail indicator. The reason that fails while taking AES tasks as a classification problems is that, obviously the range of an essay score is taken as a class. Thus as a multi- class problems, does not take the order of a class into an account which is an intrinsic property of the class labels in scoring tasks [16].

To train an AES system, a dataset of essays and their corresponding scores is typically used. The system then learns a regression model that maps the features of the essays to their scores. The model can be trained using various ML algorithms. Once the model is trained, it can be used to predict the scores of new essays. The system

extracts the relevant features from the essay and feeds them into the regression model, which produces a predicted score as model output. The predicted score can be compared to the human-assigned score to evaluate the performance of AES system. In this proposed work, AES can be seen as a regression problem, where the goal is to learn a model that predicts the scores of essays based on their features.

Types of Regression:

There are various regression algorithms that can be used to develop an AES system. The suitability of the regression algorithms depends on the complexity of the problem and the size of the dataset [17]. In this paper we have implemented ML regression types of algorithms such as MLR, Ridge Regression, Gradient Boosting Regression (GBR), Lasso Regression, Bayesian Regression (BR), Elastic Net (ELNET) Regression, Random Forest (RF) Regression, Decision Tree Regression (DT) , Gaussian process (GP) regression, Support Vector Regression (SVR), Polynomial (PR) Regression and XGBoost (Extreme Gradient Boosting) regression.

Multiple linear regression

Generally, regression helps to analyze the relationship between independent variable (x) and dependent variable (y) [18]. LR is a simple yet powerful algorithm that is easy to implement and can handle large datasets. However, it assumes that the relationship between the attributes and the score is linear, which may not always be true. The formula of the MLR is shown below:

$$y = b_0 + b_1 x_1 + b_2 x_2 + \dots + b_n x_n \quad (2)$$

where $x_1 \dots x_n$ denotes the 12 extracted features of an essay that, when combined with specific weights which is $b_0 \dots b_n$. y provide a continuous value as an final essay score. The final score is compared to the manual score, resulting in average QWK score of 0.42. The figure 1 depicts the QWK values for each essay set using MLR models.

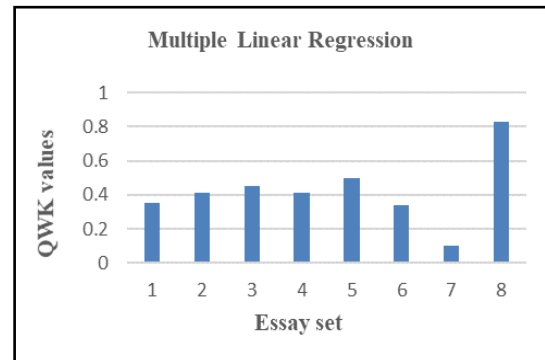


Figure: 1 QWK- Multiple linear regression

Lasso Regression

“LASSO” stands for Least Absolute Shrinkage and

Selection Operator. It is a regularization technique and deals with bias-variance trade-off problem. It is used over regression methods for a more accurate prediction and avoids the over fitting problems while there is huge difference in training and test set. The generalized formula of Lasso regression is shown in equation (2).

$$\left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\} \quad (2)$$

- p is the number of predictor variables.
- n is the number of observations.
- y_i is the response variable for the i^{th} observation.
- x_{ij} is the j^{th} predictor variable for the i^{th} observation.
- β_0 is the intercept term.
- β_j is the coefficient for the j^{th} predictor variable.
- λ is the regularization parameter.

The goal of lasso regression is to minimize the overall cost function and keeping the absolute values of the coefficients small. In our proposed AES tasks, multiple linear regression is interpretable, but remains competitive while combining the large set of variables (essay features) and those problems can also be rectified by using lasso regression [19]. After rectifying the over fitting problems and finding the optimal coefficients thus final predicted score is compared to the human score which produced an average QWK score of 0.32. The figure 2 depicts the QWK values for each essay set using lasso regression.

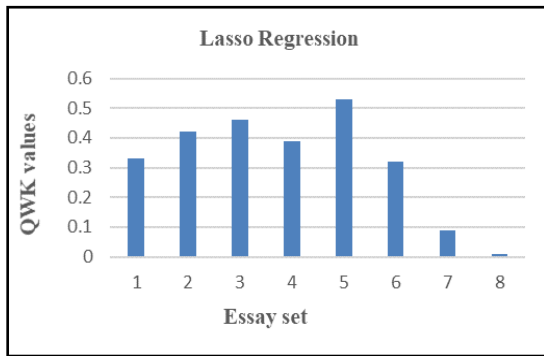


Figure: 2 QWK- Lasso regression

Ridge Regression

Ridge regression is used in AES to address the issue of over fitting, which can occur when the model is too complex and has too many features relative to the amount of data available. It can help to address over fitting by adding a penalty term to the cost function. In AES, the features used in the model are often highly correlated, which can lead to unstable coefficient estimates and poor generalization performance. Ridge regression can help to reduce the impact of multi-collinearity on the coefficient

estates and lead to more stable and accurate predictions [20]. After rectifying the over fitting problems and finding the optimal coefficients thus final predicted score is compared to the manual score and gained average QWK score of 0.32.

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \quad (3)$$

- p is the number of predictor variables.
- n is the number of observations.
- y_i is the response variable for the i^{th} observation.
- x_{ij} is the j^{th} predictor variable for the i^{th} observation.
- β_0 is the intercept term.
- β_j is the coefficient for the j^{th} predictor variable.
- λ is the regularization parameter.

The figure 3 depicts the QWK values for each essay set using ridge regression. Overall, ridge regression is a useful method for developing accurate and robust AES systems that can effectively score essays based on a variety of features. The generalized formula of ridge regression is shown in equation (3).

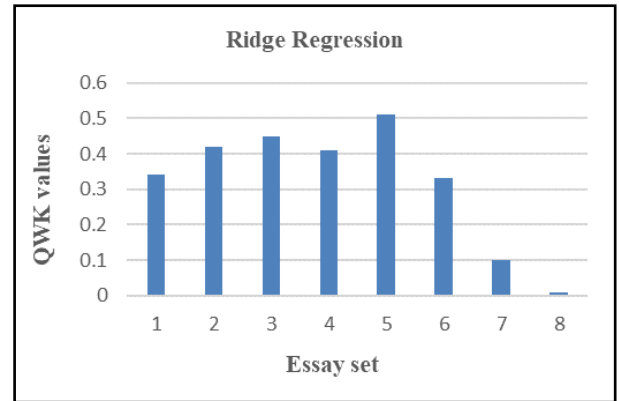


Figure: 3 QWK- Ridge regression

Gradient Boosting Regression

GBR may be used in AES assignments to estimate essay quality using a collection of characteristics. The purpose of AES is to analyze essays using ML algorithms and deliver grades comparable to those supplied by human graders. To use GBR in AES, the first step is to extract a set of features from each essay, which can include linguistic and content-based features. Once the features have been extracted, the next step is to train a gradient boosting regression model using a set of labeled training data. The model is trained to predict the scores of the essays based on their features. The training data typically consists of a set of essays that have been scored by human graders, along with the corresponding feature values [21].

$$\sum_{i=1}^N L(y_i, F_i) + \sum_{i=1}^K \Omega(f_i) \quad (4)$$

- N is the number of essays in the training set.
- $L(y_i, F_i)$ is the loss function, measuring the difference between the true score y_i and the predicted score F_i .
- K is the number of trees (weak learners) in the ensemble.
- f_i represents the individual weak learners (trees) in the ensemble.
- $\Omega(f_i)$ is a regularization term penalizing the complexity of the individual trees

Once trained, the model can be used to predict the scores of new, unlabeled essays based on their features. The predicted score can then be compared to the scores provided by human graders and achieved the average QWK score of 0.35. The figure 4 depicts the QWK values for each essay set using GBR and the generalized formula of GBR is explained in equation (4).

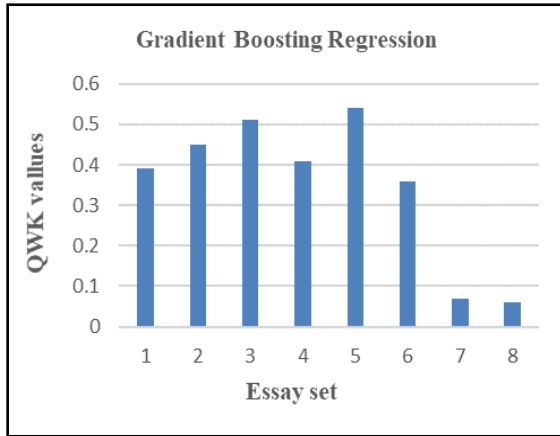


Figure: 4 QWK- Gradient Boosting regression

Bayesian Regression

The basic idea behind BR is to estimate a probability distribution over a set of parameters based on the observed data. In the context of AES, the observed data include features extracted from an essay as well as the predicted essay's score.

In this regression, prior knowledge about the parameters is combined with the observed data to form a posterior distribution. This posterior distribution can be used to estimate the probability of a given score for a new essay, given its features. The generalized the posterior distribution formula of BR algorithm is shown in equation (5). One advantage of using BR in AES is that it can incorporate prior knowledge about the distribution of scores and features. This can be particularly useful when working with small datasets or when there are particular features that are known to be important predictors of

essay quality. However, Bayesian regression can also be computationally intensive and may require significant resources to train and apply to large datasets. Additionally, the choice of prior distributions can have a significant impact on the resulting posterior distribution, and choosing appropriate priors can be a challenging task.

$$P(A|B) = \frac{P(B|A).P(A)}{P(B)} \quad (5)$$

- A, B are events.
- $P(A|B)$ is the probability of A given B is true.
- $P(B|A)$ is the probability of A given B is true.
- $P(A)$ and $P(B)$ the independent probabilities of A and B .

The author [12] proposed the Bayesian model by analyzing, extracting features and the performance of the grader is compared to the system score. Thus, the reliability and validity of the system is also investigated. Overall, Bayesian regression can be a powerful tool in AES, but its use should be carefully considered and evaluated in the context of the specific application. In our proposed work, thus final predicted score is compared to the human rater score and achieved the average QWK score of 0.32. The figure 5 depicts the QWK values for each essay set using Bayesian regression.

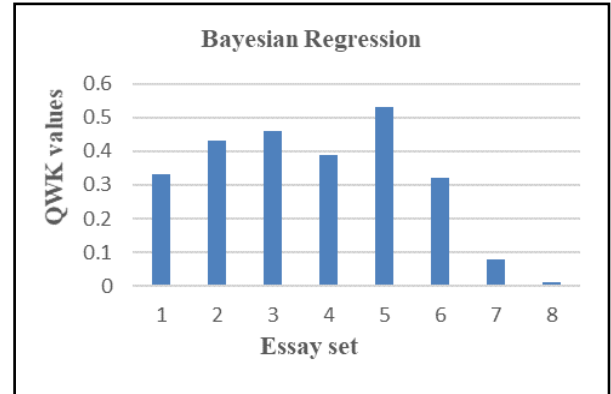


Figure: 5 QWK- Bayesian Regression

Elastic Net Regression

In the context of AES, ELNET regression can be used to predict a student's essay score on a set of features extracted from the essay. The L1 regularization term ELNET in regression can help to identify and select the most important features for predicting essay scores. This can be particularly useful when working with high-dimensional data (i.e., data with a large number of features), as it can help to reduce the risk of over fitting and improve the generalization performance of the model. The L2 regularization term sin ELNET regression helps to prevent over fitting by penalizing large coefficients in the model. This can be especially beneficial in circumstances when there are several factors that are potentially relevant predictors of essay scores, but their precise influence on

the score is ambiguous [21]. The generalized formula of ELNET is explained in equation (6).

$$J(\theta) = \sum_{i=1}^p (y_i - h_{\theta}(x_i))^2 + \lambda_1 \sum_{j=1}^q |\theta_j| + \lambda_2 \sum_{j=1}^q \theta_j^2 \quad (6)$$

- $J(\theta)$ is the objective function, where θ represents the vector of coefficients (including the intercept term).
- p is the number of training examples (essays).
- q is the number of features.
- y_i is the true score for essay i .
- $h_{\theta}(x_i)$ is the predicted score for essay i based on the current model parameters.
- λ_1 and λ_2 are the regularization parameters for the L1 and L2 penalties, respectively.

Overall, ELNET regression can be a powerful tool in AES for identifying the most important features for predicting essay scores, while also avoiding over fitting and improving the generalization performance of the model. In our proposed work, the predicted score is compared to the manual score and achieved the average QWK score of 0.01. The figure 6 depicts the QWK values for each essay set using ELNET regression model.

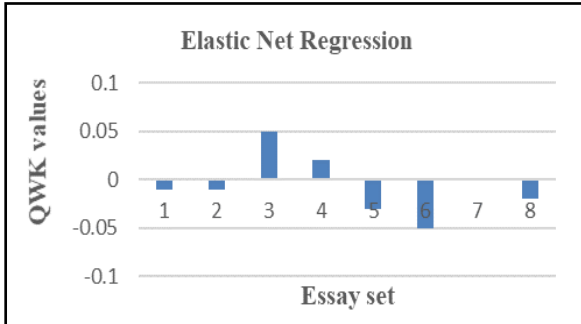


Figure: 6 QWK- Elastic Net Regression

Random Forest Regression

RF regression can also be useful in identifying the most important features for predicting essay scores. By comparing the relative relevance of each characteristic over all DT in the forest, it is feasible to determine the features that have the most influence on the final prediction. One advantage of RF regression in AES is that it can handle both categorical and numerical features, which can be useful when working with essay data that includes a mix of both types of features. Additionally, RF regression can be more robust to outliers and noisy data than some other regression techniques. The generalized formula of RF algorithm is shown in equation (8).

$$\bar{y}_i = \frac{1}{N} \sum_{j=1}^N f_j(x_i) \quad (7)$$

- y_i is the predicted score for essay i ,
- N is the number of trees in the RF.
- $f_j(x_i)$ is the prediction of the j^{th} DT for essay i .

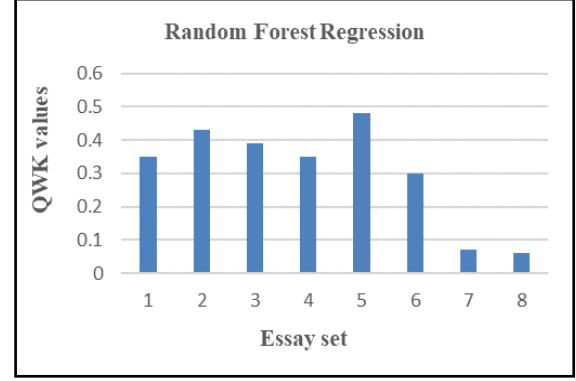


Figure: 7 QWK- Random Forest Regression

However, one potential drawback of RF regression is that the resulting model can be difficult to interpret. Because it is based on an ensemble of DTs, it can be challenging to understand exactly how each feature is contributing to the final prediction. The author proposed AES work, [23] in which thirty-five features based on readability grades, sentence information, word usage, and sentence begins and these features are sent into various ML algorithm such as RF, MLR, and SVR and thus finalize that RF performs better than other ML models. In our proposed work, the predicted score is compared to the manual score and achieved the average QWK score of 0.30. The figure 7 depicts the QWK values for each essay set using RF Regression.

Decision Tree Regression

DT regression is a ML approach that may be applied with AES. It is a sort of regression analysis that use a tree-like model to predict the value of a target variable (such as essay scores) utilizing a group of variables that predict it (such as essay features). In DT regression, the data is recursively partitioned into subsets based on the values of the predictor variables. The tree is built by selecting the best predictor variable to split the data at each step, based on a specified criterion (such as minimizing the mean squared error). The tree continues to grow until a stopping criterion is met, such as a minimum number of samples required to split a node. The generalized formula of DT regression is explained in equation (8).

$$\bar{y}_i = \frac{1}{N} \sum_{j=1}^N y_j \quad (8)$$

- y_i is the predicted score for essay i .
- N is the number of training instances that reached the leaf node during the training process.
- y_j is the true score for the j^{th} training instance.

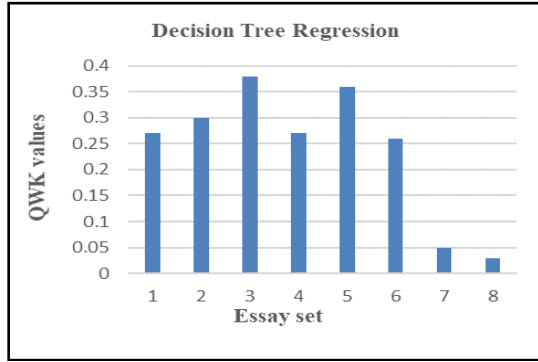


Figure: 8 QWK- Decision Tree Regression

One advantage of DT regression in AES is that it is easy to interpret and visualize the resulting model. The tree structure can provide insights into which features are the most important predictors of essay scores, and how they interact with each other. By finding the important features, the proposed model achieved the average QWK score of 0.24. The figure 8 depicts the QWK values for each essay set using DT Regression models.

Gaussian Process regression

GPR has been used in various NLP tasks, including essay scoring. In essay scoring, GPR can be used to model the relationship between various features of an essay, such as grammar, vocabulary, and organization, and its overall score. To use GPR for essay scoring, a training set of essays with their associated scores is first used to fit a GP model. The model learns the correlation between the various features and the essay scores, and the model can then be used to predict the scores of new essays.

$$f(x) \sim gp(m(x), k(x, x')) \quad (9)$$

- $f(x)$ is a random function representing the underlying latent function to be modeled.
- gp denotes a Gaussian Process.
- $m(x)$ is the mean function, representing the expected value of the function at input x .
- $k(x, x')$ is the covariance function (kernel function), representing the covariance between the function values at inputs x and x' .

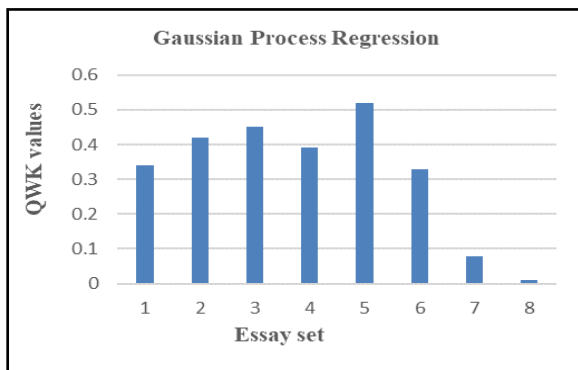


Figure: 9 QWK- Gaussian Process Regression

One advantage of GPR is that it can handle non-linear relationships between the features and the scores, which can be difficult to model using traditional MLR techniques. GPR can also handle small datasets with noise, which is common in essay scoring tasks where the number of training essays is often limited [24]. The GP model generalized formula is explained in equation (9). In our proposed work, the predicted score is compared to the manual score and achieved the average QWK score of 0.26. The figure 9 depicts the QWK values for each essay set using GP Regression.

Support vector regression

SVR is another popular non-parametric ML technique that has been applied to essay scoring tasks. Similar to GPR, SVR can handle non-linear relationships between the features and the scores, making it a suitable choice for modeling complex relationships between essays and their scores. In SVR, a training set of essays with their associated scores is used to fit a model that finds a hyper plane in a high-dimensional space that maximizes the gap between human scores and the predicted scores. The hyper plane is designed to optimize the distance between the expected scores and the hyper plane while reducing the discrepancy between the predicted and actual results. The generalized formula of SVR is explained in equation (10). The author [25] proposed using real CET4 data, where both surface level features and complex features are extracted and then fed into SVR models and yields average deviation of 0.88 when compared to human rater score.

$$f(X) = \langle w, X \rangle + b \quad (10)$$

- $f(X)$ is the predicted output.
- $\langle w, X \rangle$ denotes the dot product between the weight vector w and the input vector X .
- b is the bias term.

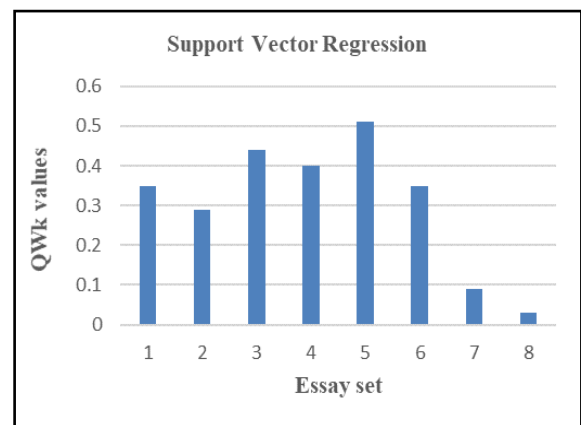


Figure: 10 QWK- Support Vector Regression

One advantage of SVR is that it can handle large datasets with high-dimensional feature spaces. It can also be more computationally efficient than GPR, making it more suitable for larger datasets. However, like GPR, SVR

requires careful selection and engineering of features that are informative and relevant to the task. Additionally, SVR may be more sensitive to parameter tuning than GPR, and the choice of kernel function can greatly affect its performance.

In our proposed work, linear and Radial Basis Function (RBF) kernel is used to get the coherence score between the human score and system generated score. In our proposed work, the predicted score is compared to the manual score and achieved the average QWK score of 0.31. The figure 10 depicts the QWK values for each essay set using SVR model.

Polynomial Regression

PR is a parametric approach to regression analysis that can also be used in essay scoring tasks. In polynomial regression, a polynomial function is used to model the relationship between the features and the essay scores. To use PR in essay scoring, a training set of essays with their associated scores is used to fit a polynomial function to the data. The degree of the polynomial function can be selected based on the complexity of the relationship between the features and the scores. The generalized formula of PR model is explained in equation (11). For example, a quadratic function (degree 2) can model a parabolic relationship between the features and the scores, while a cubic function (degree 3) can model a more complex relationship. In our proposed work, degree 3 is taken for an implementation part to get the relationship between the features of an essay and manual scores. Increasing over to degree 3 will result in over fitting problems. One advantage of PR is that it is simple to implement and interpret, making it a useful tool for understanding the relationship between the features and the scores. It can also handle non-linear relationships between the features and the scores, although it may not be as flexible as non-parametric approaches such as GPR and SVR.

$$\bar{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_n x_i^n + \varepsilon_i \quad (11)$$

- $\hat{y}_i$ is the predicted score for essay i .
- x_i is the independent variable (feature) for essay i .
- $\beta_0, \beta_1, \dots, \beta_n$ are the coefficients of the polynomial,
- n is the degree of the polynomial,
- ε_i is the error term.

However, PR can be sensitive to over fitting, especially if the degree of the polynomial function is too high [13]. It also assumes that the relationship between the features and the scores is polynomial, which may not be the case in all essay scoring tasks. But in our proposed work, the predicted score is compared to the human score and achieved the average QWK score of 0.45 (n=3).

Increasing the degree above results in over fitting issues. The figure 11 depicts the QWK values for each essay set using PR model.

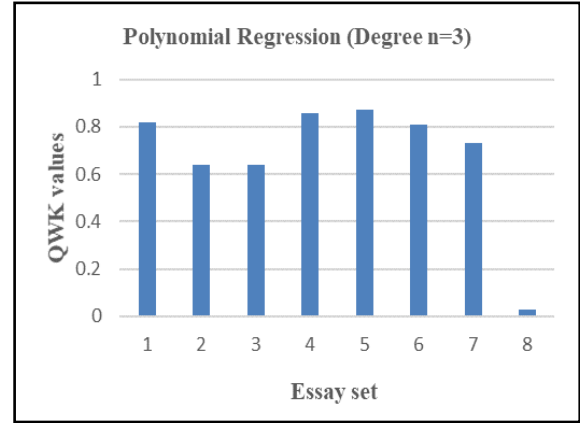


Figure: 11 QWK- Polynomial Regression (n=3)

XGBoost regression

XGBoost regression is a ML algorithm used for regression analysis based on the concept of boosting. This regression is a popular gradient boosting ML algorithm that has a set of weak regression models, such as DTs, are combined to form a strong predictive model.

$$\sum_{i=1}^N L(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (12)$$

- N is the number of essays in the training set,
- $L(y_i, \hat{y}_i)$ is the loss function measuring the difference between the true score y_i and the predicted score $\hat{y}_i$
- $\Omega(f_k)$ is a regularization term penalizing the complexity of the k -th tree.

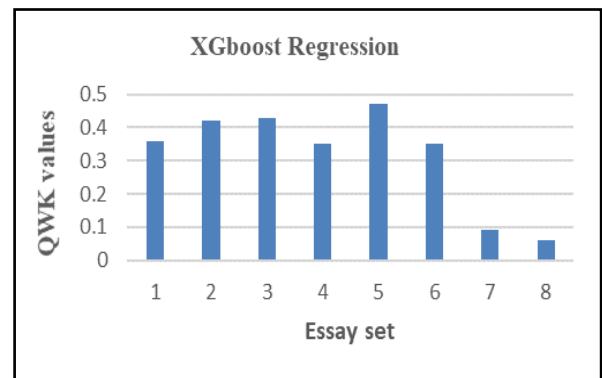


Figure: 12 QWK- XGBoost Regression

To use XGBoost in essay scoring, a training set of essays with their associated scores is used to fit a model that combines multiple DTs to predict the essay scores. The algorithm iteratively constructs DTs to lower the error between predicted and actual human scores. Each DTs is trained on a subset of the features and data points, and the

results are combined to produce a final prediction. The generalized formula of XGBOOST algorithm is shown in equation (12).

One advantage of XGBoost is that it can handle high-dimensional feature spaces and non-linear relationships between the features and the scores. It is also less prone to over fitting than some other ML techniques, as it can stop building trees once the performance on a validation set starts to decrease. In our proposed work, the human score is compared to the predicted score, resulting in average QWK score of 0.54. The figure 12 depicts the QWK values for each essay set using XGBoost Regression.

XGBoost with polynomial features regression

Performing XGBoost regression with polynomial features (XG-PR) involves extending the feature set by including polynomial combinations of the original features. This can enhance the model's capacity to capture non-linear relationships. The degree of the polynomial and other hyper parameters should be chosen carefully based on AES tasks to avoid over fitting. XGBoost algorithm into essay evaluation while also introducing polynomial features to improve the model's capacity to capture complicated relationships in the data. The generalized formula for XGBoost algorithm is shown in equation (13).

$$\hat{y} = \sum_{i=1}^N \alpha_i \cdot f_i(X) \quad (13)$$

- $\hat{y}$ is the predicted output.
- N is the number of trees in the XGBoost model.
- α_i is the weight (or coefficient) assigned to the i^{th} tree.
- $f_i(X)$ is the i^{th} tree's prediction, which is a sum of contributions from each feature.

Polynomial features up to degree d . The polynomial features can be generated by taking all combinations of features up to degree d shown in equation (14).

The polynomial features would be:

$$Poly(X, d) = [X, X^2, X^3, \dots, X^d] \quad (14)$$

In this approach, XGBoost serves as the primary regression algorithm, leveraging its gradient boosting framework to iteratively refine predictions and optimize performance. The inclusion of polynomial features involves augmenting the input features with higher-order interactions, allowing the model to better capture non-linear patterns and nuances present in the essay scoring dataset.

$$\hat{y} = \sum_{i=1}^N \alpha_i \cdot f_i(Poly(X, d)) \quad (15)$$

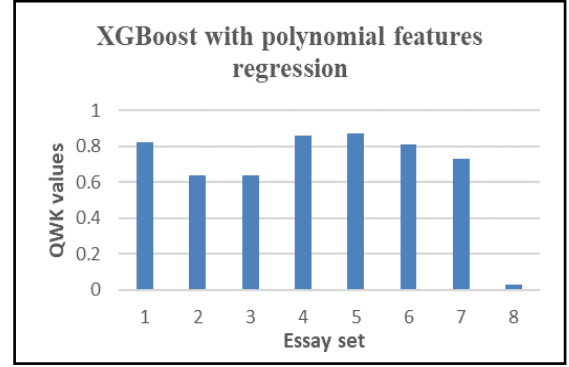


Figure: 13 QWK- XGBoost Regression with polynomial features

This approach provides a more nuanced and flexible framework for essay scoring tasks, potentially enhancing the accuracy and robustness of the predictive model. The generalized formula of XGboost regression with polynomial regression is shown in equation (15). The figure 13 depicts the QWK values for each essay set using XGBoost Regression with polynomial features and achieved an average QWK value of 0.68.

Combined Features

These above said twelve features are combined and trained using various ML algorithms. While combining among all these 12 ML models, XG-PR outperforms all other ML algorithms with an average QWK score is 0.68, which is the reason why we kept with it for our final evaluation.

Evaluation metrics

In most of the existing AES tasks, QWK method is used as a standard evaluation metrics.

Table 2 Proposed Result using ML models

ML models	Average QWK values
Multiple Linear Regression	0.42
Lasso Regression	0.32
Ridge Regression	0.31
Gradient Boosting Regression	0.35
Bayesian Regression	0.32
Elastic Net Regression	0.04
Random Forest Regression	0.30
Decision Tree Regression	0.24
Gaussian Process Regression	0.26
Support Vector Regression	0.31
Polynomial Regression	0.45
XGBOOST Regression	0.54
XGBOOST with Polynomial features	0.68

Thus, the main objective of this paper is to put forward the use of QWK as a standardized approach to assess the performance of AES systems, which can assist in

standardizing the development and evaluation of AES. Figure 14, shows the final average QWK Scores for various regression based ML models. Table 2 shows the comparison results affirm that XG-PR models gives better results than other ML models with average QWK values is 0.68. Thus XG-PR model is taken as the final predication values. Rather than the XGBoost regression with polynomial features, PR, XGBoost, PR and MLR regression also performs well with the average QWK values of 0.68, 0.54, 0.45 and 0.42 respectively.

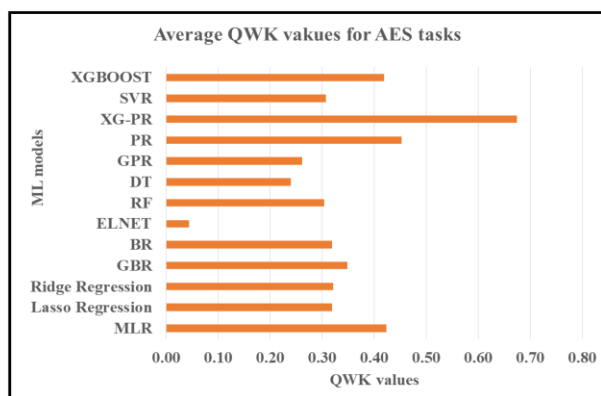


Figure: 14 ML algorithm- Average QWK score

V.CONCLUSION AND FUTURE WORK

Thus the AES is considered as an important application of ML. It helps teachers to reduce their time and cost in evaluating essays. It also helps users get a spontaneous score. In this proposed work, many ML models are used to score an essay by training and testing more than the 12,978 samples of essay in a ASAP corpus. The XGBOOST with Polynomial features was found to outperform the other ML models. The findings suggest that it is an evolving field of research, encompassing a broad range of techniques. However, it is not yet fully mature, and practical implementations are still in progress to establish a robust AES system.

FUTURE WORK

There are several ways in which the proposed work can be enhanced. The current work generates a score, based on surface level grammatical and semantic features but it can also be implemented to provide feedback along with the score in near future. The content based semantic features along with sentimental, ontological features can be enhanced in order to develop the robust AES system. Most of AES system focuses on english language but, can also be protracted to develop the automated essay score in low resource languages.

REFERENCES:

[1] K. Lee, M. Fanguy, B. Bligh, and X. S. Lu, "Adoption of online teaching during the COVID-19 Pandemic: a systematic analysis of changes in university teaching activity," *Educ. Rev.*, vol. 74, no. 3, pp. 460–483, Apr. 2022, doi: 10.1080/00131911.2021.1978401.

[2] T. Yambi and C. Yambi, "ASSESSMENT AND EVALUATION IN EDUCATION," Jul. 2020.

[3] D. Ramesh and S. K. Sanampudi, "An automated essay scoring systems: a systematic literature review," *Artif. Intell. Rev.*, vol. 55, no. 3, pp. 2495–2527, Mar. 2022, doi: 10.1007/s10462-021-10068-2.

[4] K. A. Osakwe, K. Ola, and P. Omotosho, "Automated Essay Scoring (AES) System in COVID 19 Pandemic: A Contactless Academia Approach," *Res. Highlights Lang. Lit. Educ.* Vol 4, pp. 1–20, Apr. 2023, doi: 10.9734/bpi/rhlle/v4/3504A.

[5] "Advances in the Field of Automated Essay Evaluation - Google Search," <https://www.google.com/search?q=Advances+in+the+Field+of+Automated+Essay+Evaluation&sourceid=chrome&ie=UTF-8> (accessed May 08, 2023).

[6] "Dikli (2006) Dikli S. An overview of automated scoring of essays. The Journal Of Technology, Learning, and Assessment. 2006;5(1):1–36. - Google Search," [https://www.google.com/search?q=Dikli+\(2006\)+Dikli+S.+An+overview+of+automated+scoring+of+essays.+The+Journal+Of+Technology%2C+Learning%2C+and+Assessment.+2006%3B5\(1\)%3A1%E2%80%9336.&sourceid=chrome&ie=UTF-8](https://www.google.com/search?q=Dikli+(2006)+Dikli+S.+An+overview+of+automated+scoring+of+essays.+The+Journal+Of+Technology%2C+Learning%2C+and+Assessment.+2006%3B5(1)%3A1%E2%80%9336.&sourceid=chrome&ie=UTF-8) (accessed May 08, 2023).

[7] T. K. Landauer, "Automatic Essay Assessment," *Assess. Educ. Princ. Policy Pract.*, vol. 10, no. 3, pp. 295–308, Nov. 2003, doi: 10.1080/0969594032000148154.

[8] C. Ramineni and D. Williamson, "Understanding Mean Score Differences Between the e-rater® Automated Scoring Engine and Humans for Demographically Based Groups in the GRE® General Test," *ETS Res. Rep. Ser.*, vol. 2018, no. 1, pp. 1–31, 2018, doi: 10.1002/ets2.12192.

[9] M. Mahana, M. Johns, and A. Apte, "Automated Essay Grading Using Machine Learning".

[10] J. Chen, J. H. Fife, I. I. Bejar, and A. A. Rupp, "Building e-rater® Scoring Models Using Machine Learning Methods," *ETS Res. Rep. Ser.*, vol. 2016, no. 1, pp. 1–12, 2016, doi: 10.1002/ets2.12094.

[11] "Exploring the potential of using an AI language model for automated essay scoring | Elsevier Enhanced Reader," <https://reader.elsevier.com/reader/sd/pii/S2772766123000101?token=74BD56B19A47A1CEE4849C3C64947E27D2084CE679032AF1DABE947329ACA5706C4AC273ECB50C22136B612E5AB9CEFD&originRegion=eu-west-1&originCreation=20230510201051> (accessed May 11, 2023).

[12] A. Singh and D. Pant, "Automated essay scoring using machine learning," *Int. J. Adv. Res. Ideas Innov. Technol.*, vol. 5, no. 2, pp. 1154–1156, Jun. 2019, Accessed: Mar. 11, 2023. [Online]. Available: <https://www.ijariit.com/manuscript/automated-essay-scoring-using-machine-learning/>

[13] "Automated English Digital Essay Grader Using Machine Learning - Google Search," <https://www.google.com/search?q=Automated+English+Digital+Essay+Grader+Using+Machine+Learning&sourceid=chrome&ie=UTF-8> (accessed May 11, 2023).

[14] A. C. Fong, "Finding the Depth of a Flow Graph".

[15] T. Zesch, M. Wojatzki, and D. Scholten-Akoun, "Task-Independent Features for Automated Essay Grading," in *Proceedings of the Tenth Workshop on Innovative Use of NLP for Building Educational Applications*, Denver, Colorado: Association for Computational Linguistics, Jun. 2015, pp. 224–232, doi: 10.3115/v1/W15-0626.

[16] "Regression or classification? Automated Essay Scoring for Norwegian - Google Search," <https://www.google.com/search?q=Regression+or+classification%3F+Automated+Essay+Scoring+for+Norwegian&sourceid=chrome&ie=UTF-8> (accessed May 11, 2023).

[17] "first_page settings Order Article Reprints Open AccessArticle Twenty Years of Machine-Learning-Based Text Classification: A Systematic Review - Google Search."

[18] D. Maulud and A. M. Abdulazeez, "A Review on Linear Regression Comprehensive in Machine Learning," *J. Appl. Sci.*

- Technol. Trends, vol. 1, no. 4, Art. no. 4, Dec. 2020, doi: 10.38094/jastt1457.
- [19] S. Somasundaran, C. M. Lee, M. Chodorow, and X. Wang, "Automated Scoring of Picture-based Story Narration," in Proceedings of the Tenth Workshop on Innovative Use of NLP for Building Educational Applications, Denver, Colorado: Association for Computational Linguistics, Jun. 2015, pp. 42–48. doi: 10.3115/v1/W15-0605.
- [20] A. Bager, M. Roman, M. Algelidh, and B. Mohammed, "Addressing Multicollinearity In Regression Models: A Ridge Regression Application," J. Soc. Econ. Stat., vol. 6, no. 1, pp. 30–45, 2017, Accessed: May 11, 2023. [Online]. Available: <https://ideas.repec.org//a/aes/jsesro/v6y2017i1p30-45.html>
- [21] "Regression analysis using gradient boosting regression tree: Aurora articles | NEC." <https://www.nec.com/en/global/solutions/hpc/articles/tech14.html> (accessed May 11, 2023).
- [22] V. Johnson, "On Bayesian analysis of multirater ordinal data: An application to automated essay grading," J. Am. Stat. Assoc., vol. 91, pp. 42–51, Mar. 1996, doi: 10.2307/2291381.
- [23] H. Ghanta, "Automated Essay Evaluation Using Natural Language Processing and Machine Learning".
- [24] D. E. Beck, "Gaussian Processes for Text Regression".
- [25] Y. Li and Y. Yan, "An Effective Automated Essay Scoring System Using Support Vector Regression," 2012 Fifth Int. Conf. Intell. Comput. Technol. Autom., pp. 65–68, Jan. 2012, doi: 10.1109/ICICTA.2012.23.
- [26] "A Study of Polynomial Regression towards Machine Learning - Google Search." <https://www.google.com/search?q=A+Study+of+Polynomial+Regression+towards+Machine+Learning&sourceid=chrome&ie=UTF-8> (accessed May 12, 2023).